

Abhishek P G

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Profile

Physics undergraduate with a specialization in AI & Data Science, interested in quantum information, AMO physics, photonics, and computational physics. Experienced in numerical modelling and data analysis, with hands-on work in finite difference methods, analytical techniques, and statistical validation. Comfortable working across Python, R, and MATLAB. Looking for research opportunities where rigorous computation meets fundamental physics.

Research Experience

IEEE EMBS Summer Internship 2026

(Selected)

IIT Kharagpur - Remote

- Selected as a Research Intern for the IEEE EMBS Student Branch Chapter, IIT Kharagpur Summer Internship 2026.
- Working on a project involving **Finite Element Modelling (FEM)**.
- Utilizing numerical methods and computational techniques for engineering simulations..

Summer Internship Fellowship 2026

June – July 2026

Quantum Information, SRM University AP, Andhra Pradesh

Guide: Dr. Soumyakanti Bose

- Studied Quantum Error Correction (QEC), focusing on 3-qubit bit-flip, phase-flip, and Shor's 9-qubit codes.
- Explored stabilizer formalism and syndrome-based error correction.
- Authored a technical report on the mathematical foundations of QEC.

Summer Internship Program

May 2026

Department of Physics, Sacred Heart College (Autonomous), Chalakudy

Guide: Dr. Vijitha Mukundan

- Studied the 1D heat equation using analytical and numerical methods, including Fourier series, Crank–Nicolson, and Gauss–Seidel iteration.
- Implemented and validated numerical solutions by analysing convergence, stability, and accuracy against analytical results.
- Documented the mathematical formulation, computational methodology, and results in a technical report.

REYES Mentorship Programme

2025

University of California, Berkeley (Remote)

Mentor: Satchith P. S., Amrita Vishwa Vidyapeetham, Coimbatore

- Co-authored "Foundations of Photonic Quantum Computing: A Pedagogical Introduction," covering linear optical quantum computing, photonic qubit encodings, entanglement, and quantum algorithms
- Developed educational content on polarization, path, time-bin, frequency, and OAM encodings and their roles in quantum information processing
- Conducted literature review across quantum optics and quantum information, synthesising contemporary research into an undergraduate-accessible format

Projects

IRNetra AI — Structure-Preserving Enhancement and Pseudo-Colorization of Infrared Satellite Images 2026

Bharatiya Antariksh Hackathon (BAH) 2026

- Co-conceptualized a hybrid pipeline integrating classical image processing techniques (OpenCV: CLAHE, denoising, edge enhancement) with a proposed Pix2Pix Conditional GAN framework for structure-preserving pseudo-colorization of monochromatic infrared satellite imagery.
- Designed the conceptual workflow targeting applications in disaster management, border surveillance, environmental monitoring, precision agriculture, and search-and-rescue operations.
- Collaborated with a four-member interdisciplinary team (Physics, Chemistry, and Mathematics) to develop the technical concept note, methodology, and hackathon submission.

Education

B.Sc. Physics (Specialization in AI & Data Science) 2024 – Present

St. Aloysius College, Elthuruth (University of Calicut)

- CGPA: 8.46
- Minor in Mathematics: Calculus, Linear Algebra, Differential Equations, Fourier Analysis
- Minor in Statistics: Probability Theory, Inferential Statistics, Data Science with R

Senior Secondary (Class XII), CBSE — 80% 2022

Ansar English School, Perumpilavu

Secondary School (Class X), KBPE — Full A+ 2020

Bethany St. John's EHSS, Kunnankulam

Research Interests

Quantum Information and Quantum Computation • Quantum Error Correction • Quantum Optics and Photonics • Computational Physics • Artificial Intelligence for Scientific Applications

Technical Skills

Programming Python (NumPy, SciPy, Matplotlib, Qiskit), R, MySQL

Scientific MATLAB/Octave, Anaconda, Jupyter Notebook

Productivity L^AT_EX, Overleaf, MS Office, LibreOffice, Canva

Workshops & Conferences

- Quantum Information and Emerging Quantum Technologies (QIEQT) – Indian Institute of Technology (IIT) Rourkela (2026)
- Quantum Mechanics to Computation (QM2C) – SRM University-AP (2026)
- Master's Colloquium – International Conference on Operators and Spectral Theory (ICOST 2025) – Carmel College (Autonomous), Mala (2025)
- Workshop on Introduction to Quantum Computation & Information – SARBTM Government College, Koyilandy (2025)
- Physics in Action: Emerging Trends Workshop – Carmel College (Autonomous), Mala (2025)

Achievements & Leadership

- Best Student Award — Don Bosco Trust Award, St. Aloysius College (2026)
- First Prize — Intercollegiate Mathematics Quiz, Helix (2026)
- First Prize — All Kerala Science Quiz, EKN, Christ College Irinjalakkuda (2024)
- First Prize — Oral Presentation: Gravitational Waves, St. Thomas College Thrissur (2024)
- Third Prize — Oral Presentation: Linear Optical Quantum Computing, Christ College Irinjalakkuda (2025)
- Fourth Place — All Kerala Intercollegiate Physics Quiz, MG University Campus (2024)
- Student Coordinator — Alpha Physics Club, St. Aloysius College
- Volunteer & Coordinator — All Kerala Physics Fest ENIGMA 2026, St. Aloysius College
- NCC Cadet — Leadership and discipline training (2024–Present)

Languages: English (Fluent) · Malayalam (Native) · Hindi (Proficient)

References available upon request.